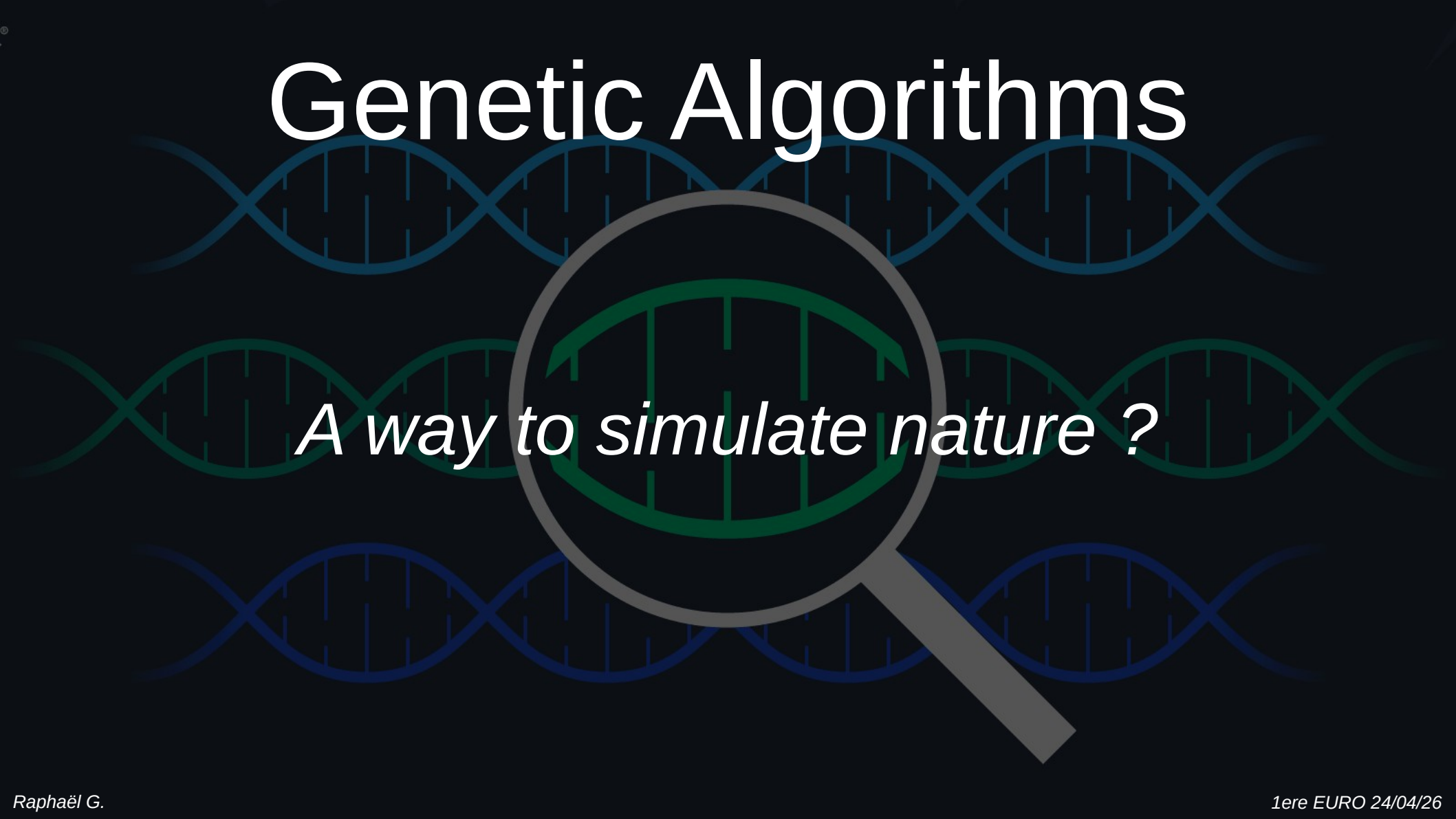


Genetic Algorithms



A way to simulate nature ?

Introduction : GA

- Metaheuristic : good but not optimal
- Belongs to Evolutionary Algorithms (EA)
- Classical methods : Gradient || Exact



$$f'(x_0^-) \neq f'(x_0^+)$$

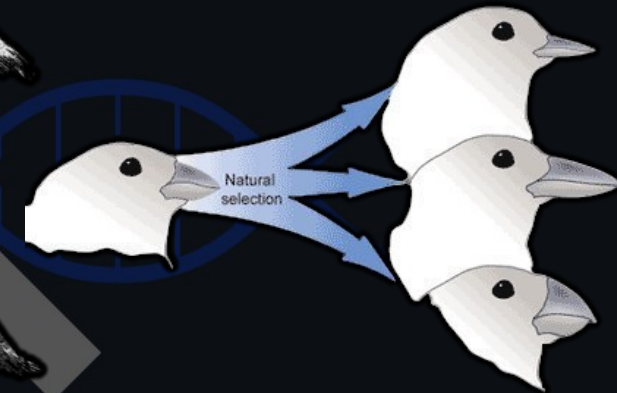
$$\binom{n}{k}$$

$$n!$$

Natural Selection

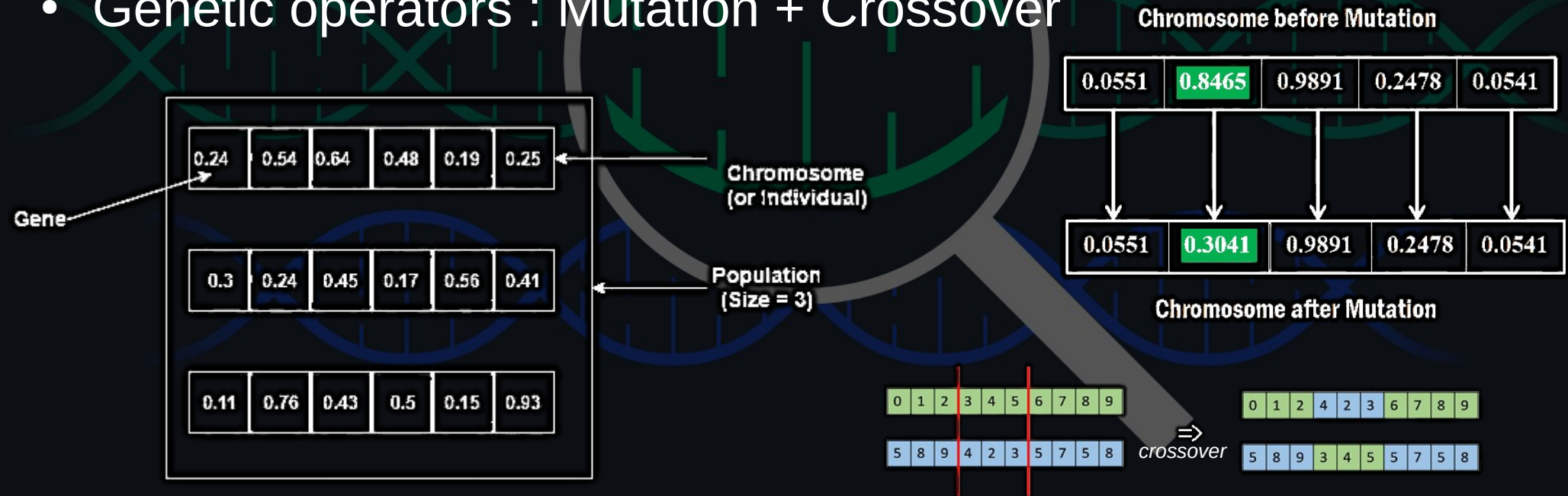
- 1859 : Advantages $\Rightarrow$ +Survival $\Rightarrow$ +Reproduction $\Rightarrow$ + Allele frequency
- Finch in the Galapagos Islands
- Bigger $\Rightarrow$ Big seeds || Smaller $\Rightarrow$ Small Insects || Longer $\Rightarrow$ Cactus nectar

THE THEORY OF EVOLUTION



GA : Presentation

- Population : many individuals
- Chromosome : array of genes
- Genetic operators : Mutation + Crossover



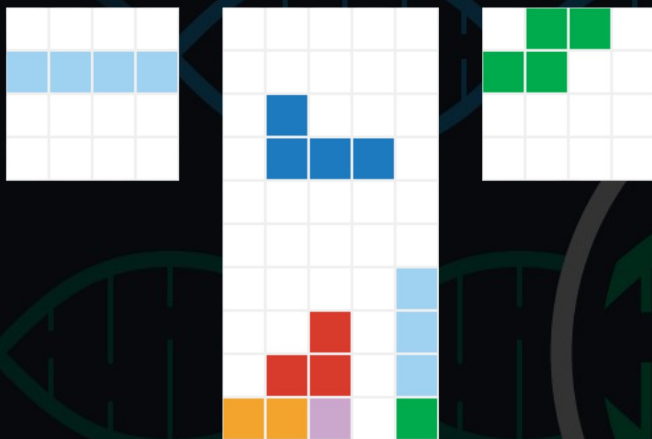
GA : Basic Concept

```
def GA(nbGenerations, nbGames):  
    for iGen in range(nbGenerations):  
        '''Evaluation'''  
        for iGa in range(nbGames):  
            for iI in range(self.nbIndividuals):  
                self.play_until_die(self.lIndividuals[iI])  
            '''Genetic operators'''  
            self.lIndividuals.sort(key=lambda individual: individual.score)  
            for iSu in range(self.nbSurvivors):  
                start = min(self.nbIndividuals - 1, self.nbSurvivors + iSu * self.nbMutated)  
                for iMu in range(start, start + self.nbMutated):  
                    self.mutate(self.lIndividuals[iMu], self.lIndividuals[iSu])  
  
function GA:  
    for each generation:  
        '''Evaluation'''  
        for each game:  
            for each individual:  
                play_until_die()  
        '''Genetic operators'''  
        sort_individuals_by_score()  
        keep_top_30%_unchange()  
        modifie_others_with_mutations()
```

Limitations and Improvements

Limitations	Improvements
Real creatures too complex	Give more observations + Give them a bigger 'brain'
Linear operations for score	Non-linear operations => better 'brain'
Fall into a bad local optimum	Global research phase
Real evolutionary process very complex	Combine mutations + crossovers
Slow simulations (fitness function)	None

GA example : Tetris bots

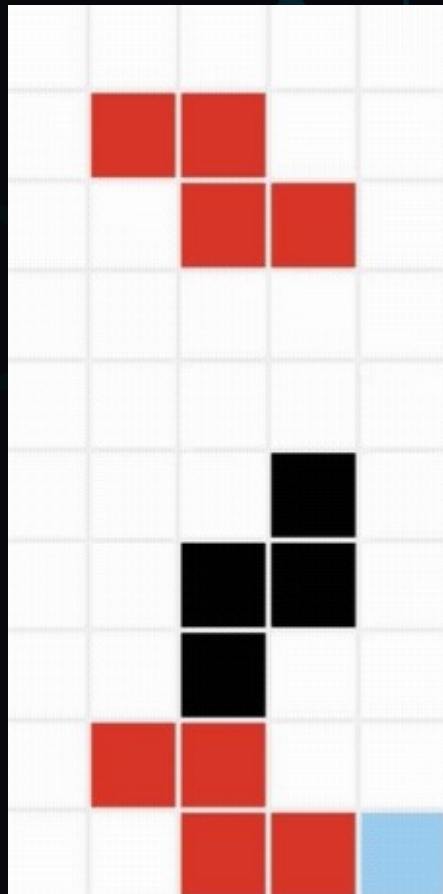


POPULATION OF 30 MINI TETRIS INDIVIDUAL PLAYERS

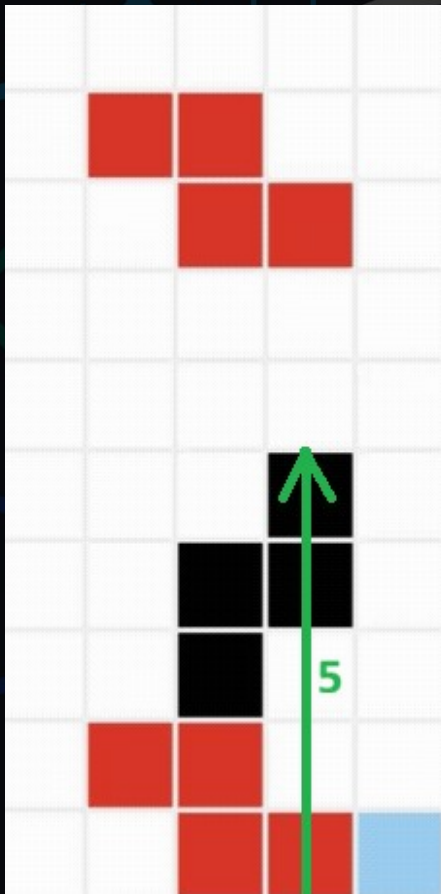


Tetris : Observations

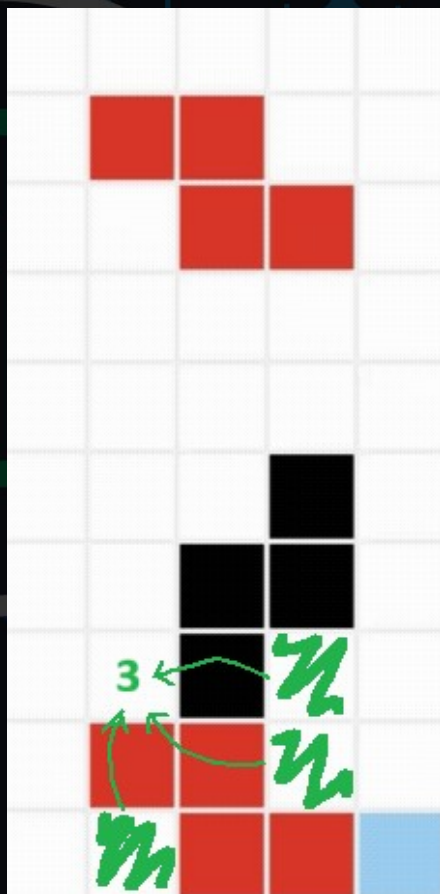
Grid



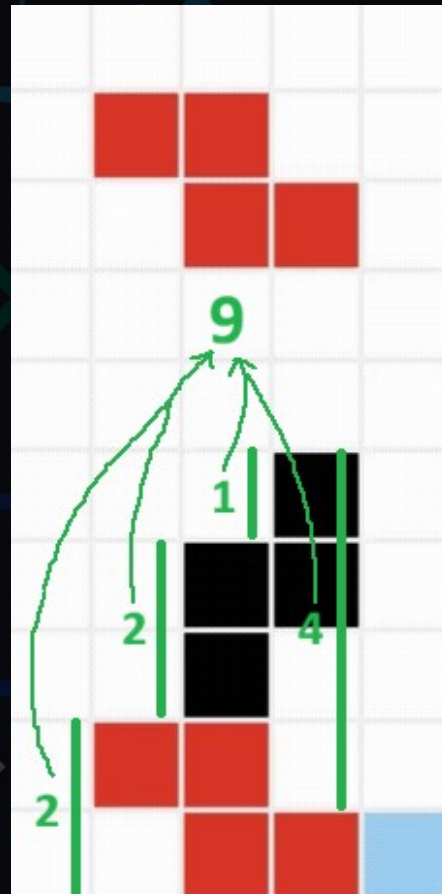
Max Height Grid



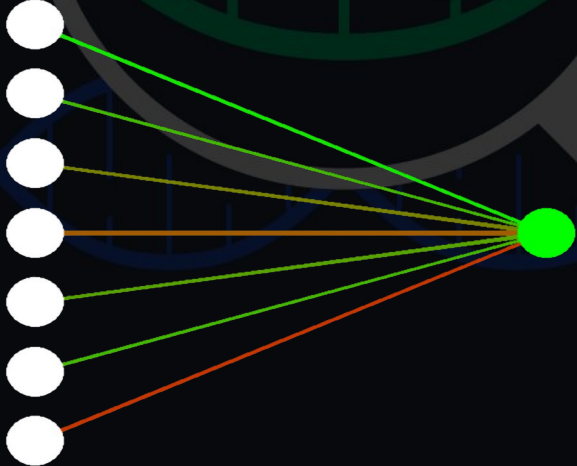
Number Holes



Flatness



Tetris : Score function

$$\begin{bmatrix} 0.137 \\ 0.842 \\ 0.509 \\ 0.276 \\ 0.931 \\ 0.664 \\ 0.415 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \\ w_7 \end{bmatrix}^T \begin{bmatrix} \text{value}_{\text{maxHeightGrid}} \\ \text{value}_{\text{maxHeightPiece}} \\ \text{value}_{\text{sumHeights}} \\ \text{value}_{\text{nbHoles}} \\ \text{value}_{\text{flatness}} \\ \text{value}_{\text{nbCompleteLines}} \\ \text{value}_{\text{nbWells}} \end{bmatrix} = \begin{bmatrix} -0.742 \\ 0.315 \\ -0.128 \\ 0.904 \\ -0.557 \\ 0.221 \\ -0.873 \end{bmatrix}^T \begin{bmatrix} \text{value}_{\text{maxHeightGrid}} \\ \text{value}_{\text{maxHeightPiece}} \\ \text{value}_{\text{sumHeights}} \\ \text{value}_{\text{nbHoles}} \\ \text{value}_{\text{flatness}} \\ \text{value}_{\text{nbCompleteLines}} \\ \text{value}_{\text{nbWells}} \end{bmatrix}$$


- Non-differentiable $f'(x_0^-) \neq f'(x_0^+)$
- Combinatorial $21^7 \approx 18014398509481984$

Tetris : Genetic operators

POPULATION OF 30 MINI TETRIS INDIVIDUAL PLAYERS


$$\begin{bmatrix} 0.523 \\ -0.781 \\ 0.104 \\ -0.332 \\ 0.999 \\ -0.457 \\ 0.210 \end{bmatrix}$$

mutation $a + a \cdot \mathcal{N}(0,1) \cdot 0.1$

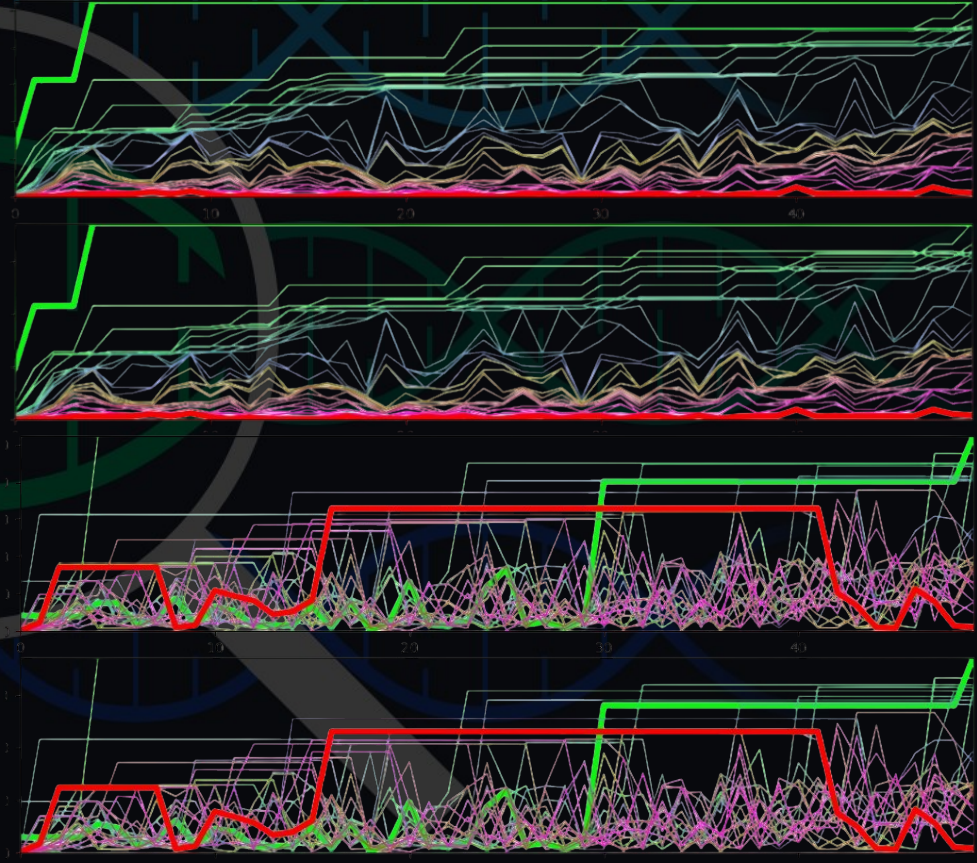
$$\begin{bmatrix} 0.548 \\ -0.720 \\ 0.095 \\ -0.360 \\ 1.042 \\ -0.430 \\ 0.198 \end{bmatrix}$$

- Score on 1 game
- 5x10 grid : eventually die
- Mutation rate 0.15 $\rightsquigarrow$ 0.01

Simulation 1

- => 1h30
- => 28k score + 814 moves

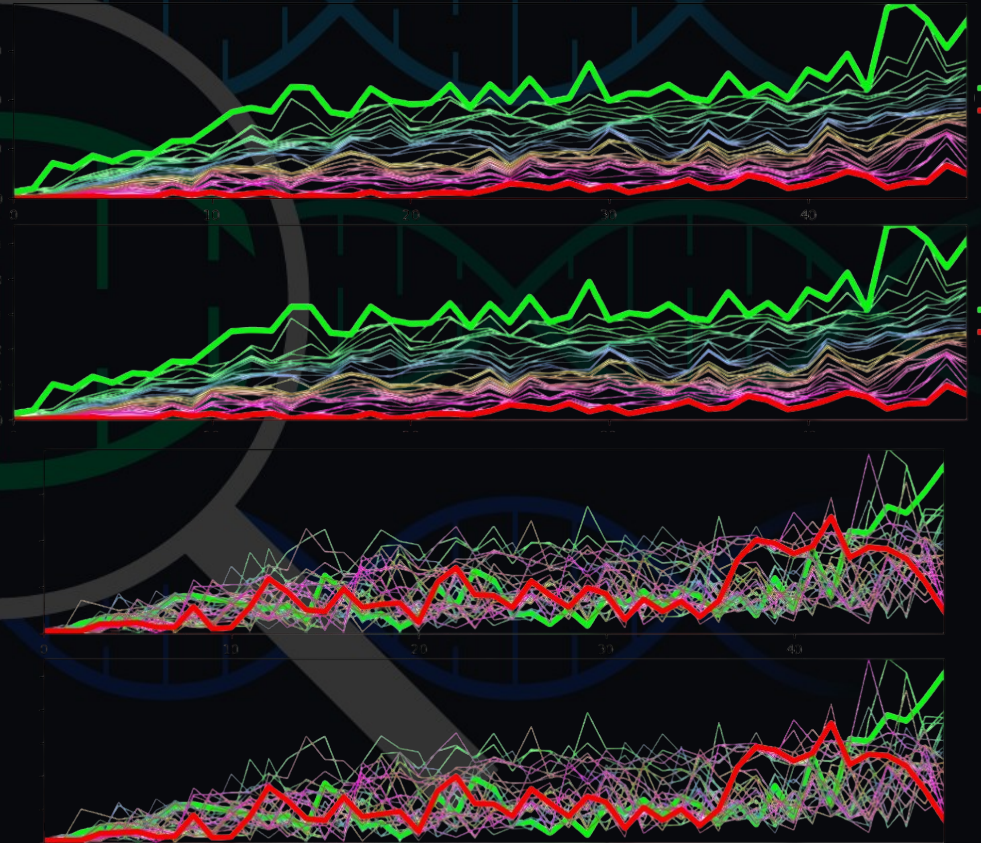
$$S = \begin{bmatrix} -0.23 \\ 0.21 \\ -0.69 \\ -0.40 \\ -0.37 \\ 1.02 \\ -0.52 \end{bmatrix} \cdot \begin{bmatrix} \text{value}_{\text{maxHeightGrid}} \\ \text{value}_{\text{maxHeightPiece}} \\ \text{value}_{\text{sumHeights}} \\ \text{value}_{\text{nbHoles}} \\ \text{value}_{\text{flatness}} \\ \text{value}_{\text{nbCompleteLines}} \\ \text{value}_{\text{nbWells}} \end{bmatrix}$$



Simulation 2

- => 7h
- => 55k score + 1600 moves

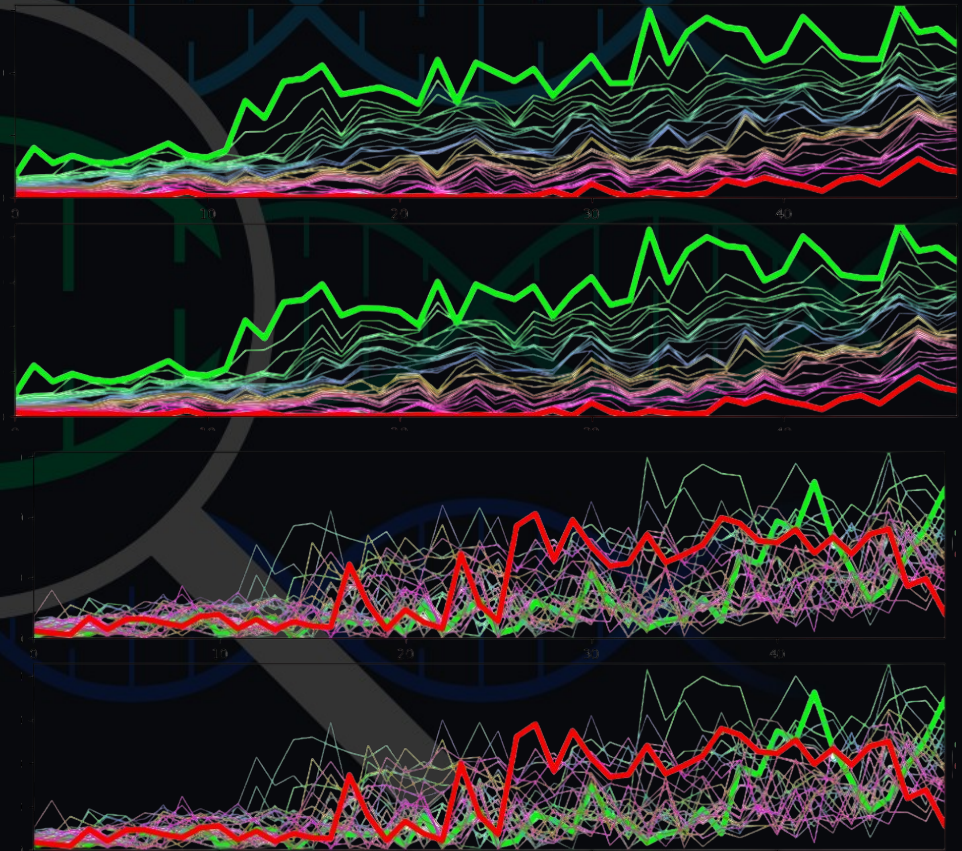
$$S = \begin{bmatrix} -0.01 \\ -0.15 \\ 0.04 \\ -0.92 \\ -0.03 \\ 0.83 \\ -1.13 \end{bmatrix} \cdot \begin{bmatrix} \text{value}_{\text{maxHeightGrid}} \\ \text{value}_{\text{maxHeightPiece}} \\ \text{value}_{\text{sumHeights}} \\ \text{value}_{\text{nbHoles}} \\ \text{value}_{\text{flatness}} \\ \text{value}_{\text{nbCompleteLines}} \\ \text{value}_{\text{nbWells}} \end{bmatrix}$$



Simulation 3

- => 12h30
- => 115k score + 3250 moves

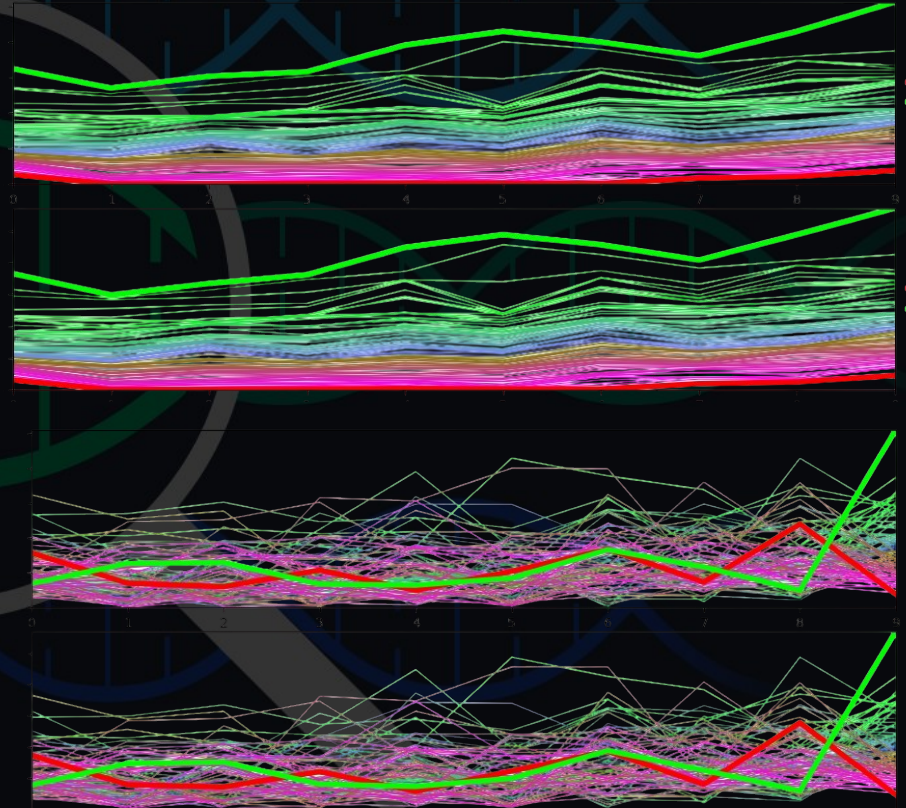
$$S = \begin{bmatrix} 0.26 \\ -0.36 \\ -0.58 \\ -0.76 \\ -0.17 \\ -0.12 \\ -1.39 \end{bmatrix} \cdot \begin{bmatrix} \text{value}_{\text{maxHeightGrid}} \\ \text{value}_{\text{maxHeightPiece}} \\ \text{value}_{\text{sumHeights}} \\ \text{value}_{\text{nbHoles}} \\ \text{value}_{\text{flatness}} \\ \text{value}_{\text{nbCompleteLines}} \\ \text{value}_{\text{nbWells}} \end{bmatrix}$$



Simulation 4

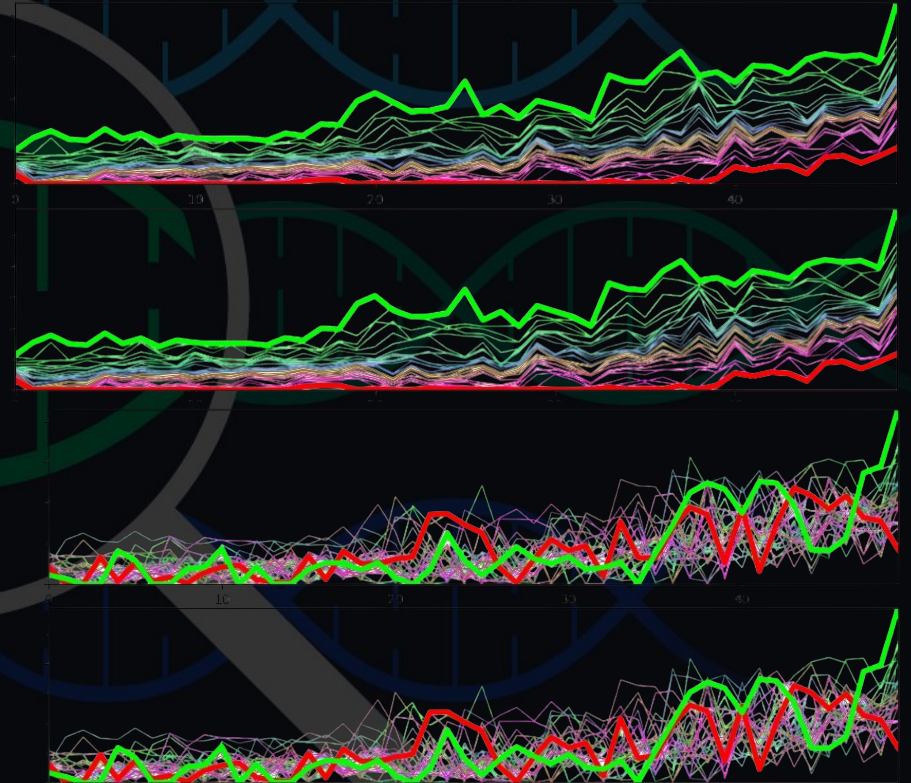
- => 8h40
- => 111k score + 3150 moves

$$S = \begin{bmatrix} 0.22 \\ -0.26 \\ -0.07 \\ -0.86 \\ -0.16 \\ 0.07 \\ -0.99 \end{bmatrix} \cdot \begin{bmatrix} \text{value}_{\text{maxHeightGrid}} \\ \text{value}_{\text{maxHeightPiece}} \\ \text{value}_{\text{sumHeights}} \\ \text{value}_{\text{nbHoles}} \\ \text{value}_{\text{flatness}} \\ \text{value}_{\text{nbCompleteLines}} \\ \text{value}_{\text{nbWells}} \end{bmatrix}$$



Simulation 5

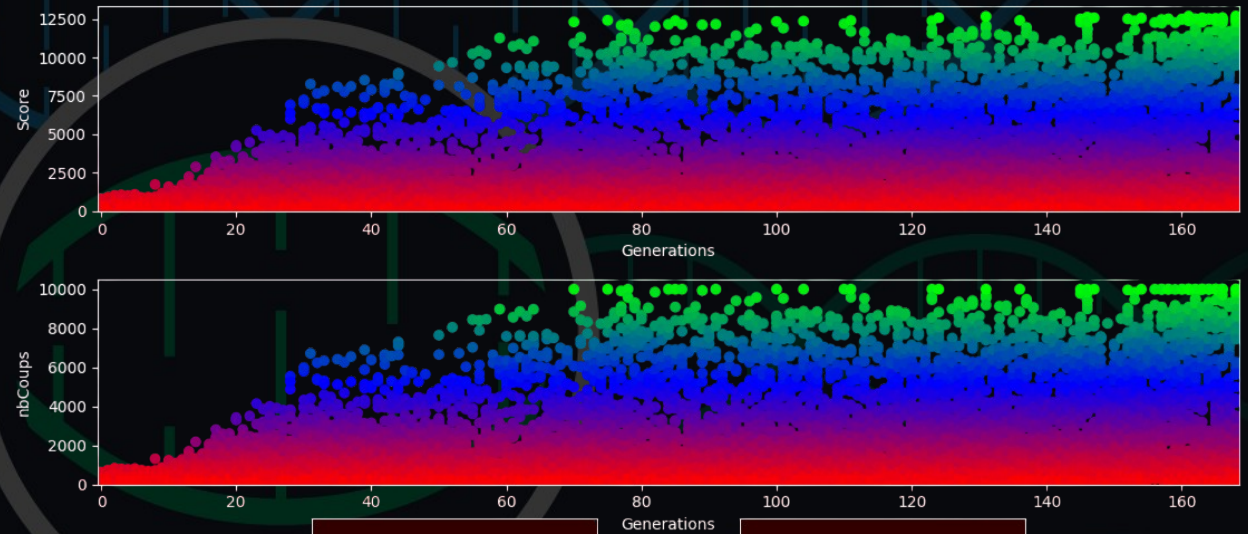
- => 9h30
- => 97k score + 2600 moves



GA : Game 2

Mode 1 ?

100 Joueurs || 3 Balles || 169 nbGen || 5 nbParties || 11→16→3 || 2 kNN



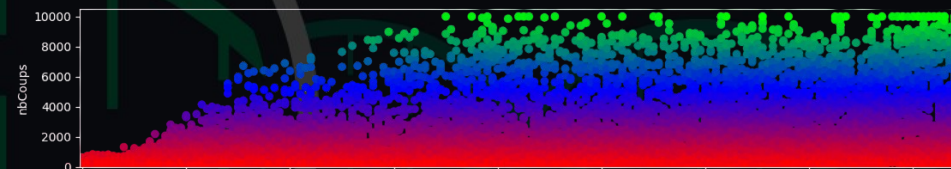
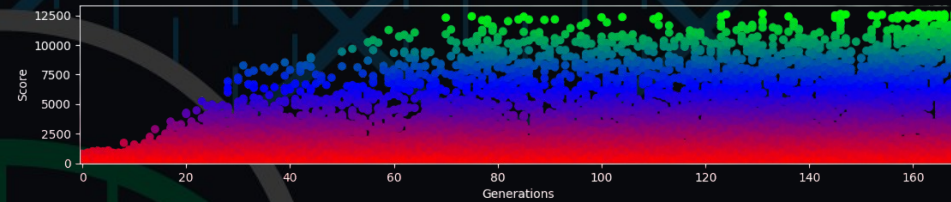
Avec animations ?

Normal ?

Simulation 2

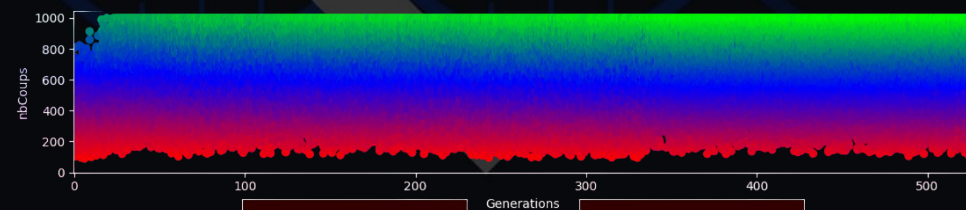
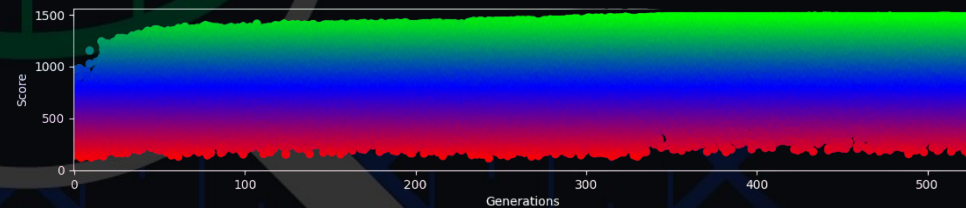
Mode 1 ?

100 Joueurs || 3 Balles || 169 nbGen || 5 nbParties || 11→16→3 || 2 kNN



Mode 1 ?

100 Joueurs || 3 Balles || 527 nbGen || 10 nbParties || 11→16→3 || 2 kNN

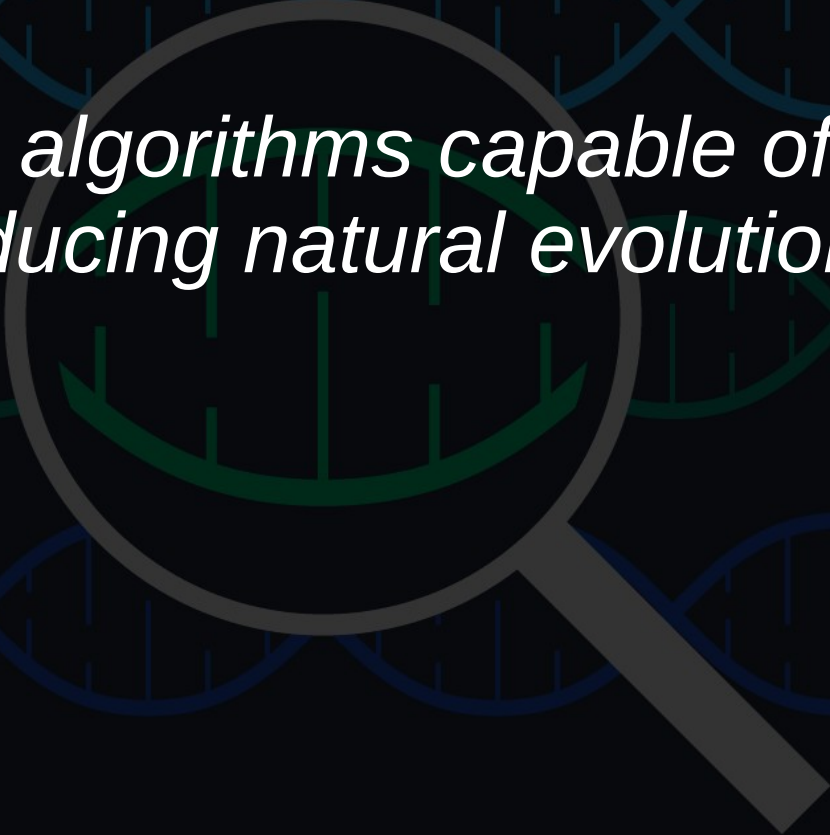


Avec animations ?

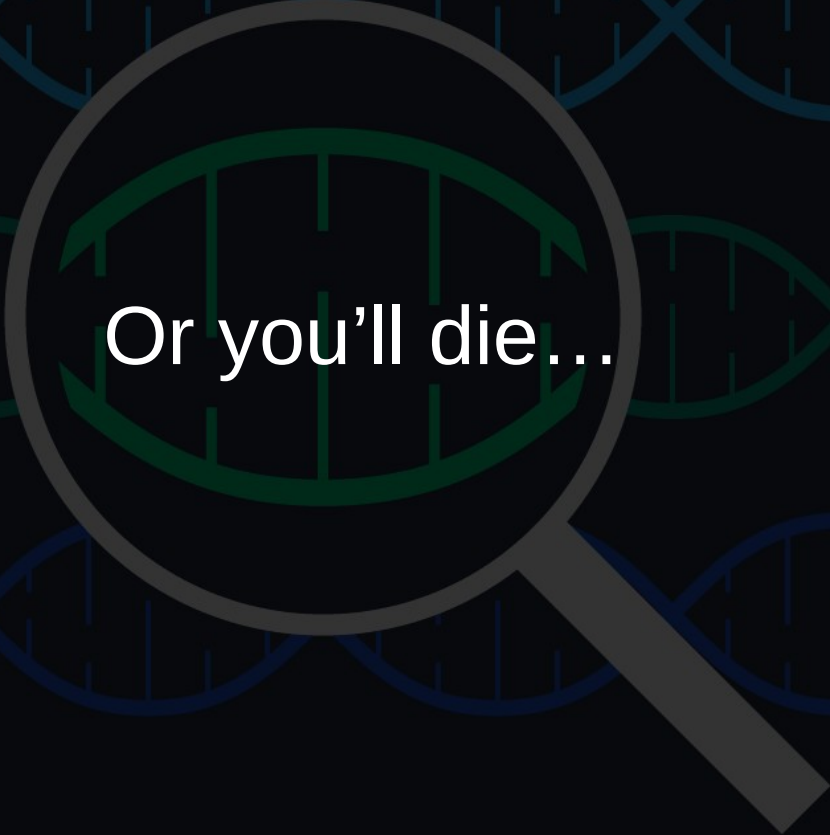
Normal ?

Conclusion

Are genetic algorithms capable of faithfully reproducing natural evolution ?

A magnifying glass with a grey handle and frame is positioned over the text. The background features a repeating pattern of DNA double helices in shades of blue and green.

DO NOT GO ANY FURTHER !!!

A magnifying glass with a grey handle and frame is positioned over a DNA double helix. The DNA structure is rendered in three colors: a light blue helix at the top, a dark green helix in the middle, and a medium blue helix at the bottom. The magnifying glass is focused on the dark green helix, which is the central element of the image. The text "Or you'll die..." is written in white inside the lens of the magnifying glass.

Or you'll die...

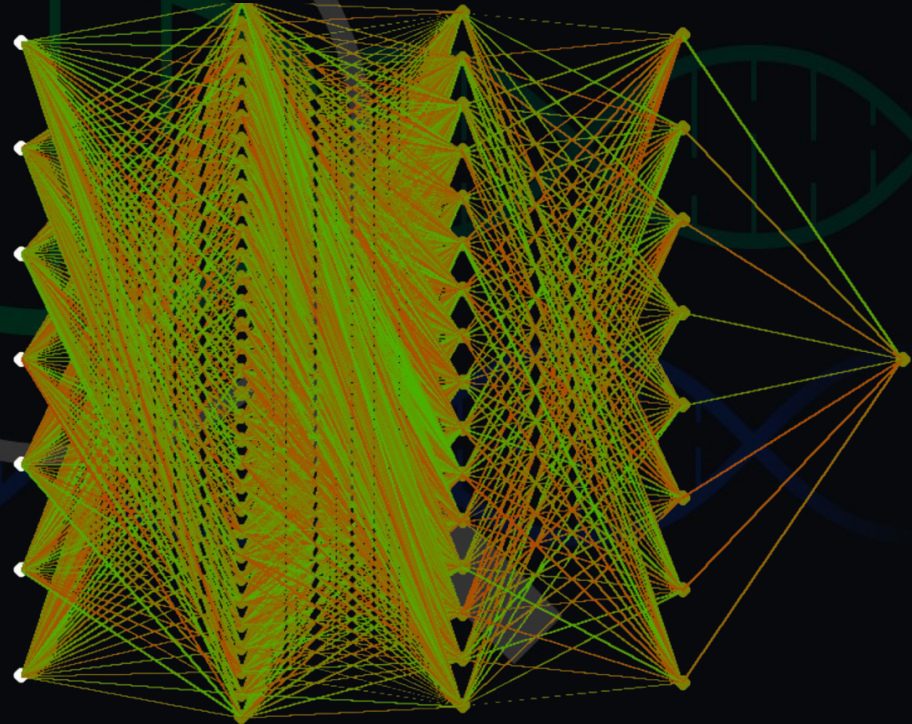
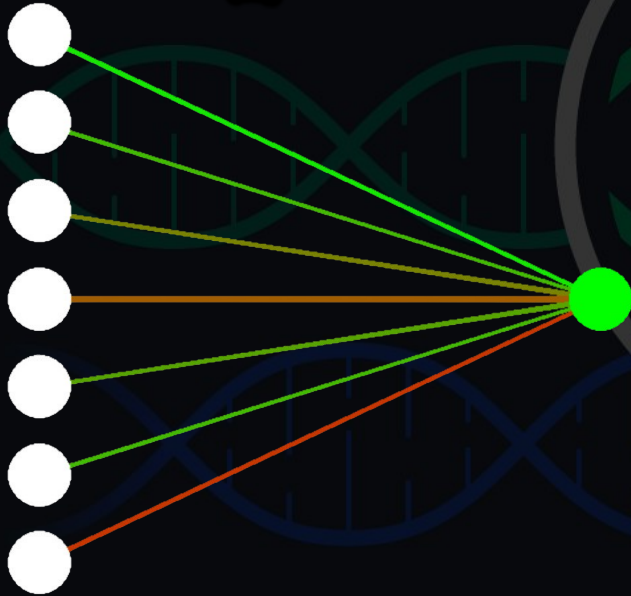
YOU DIED



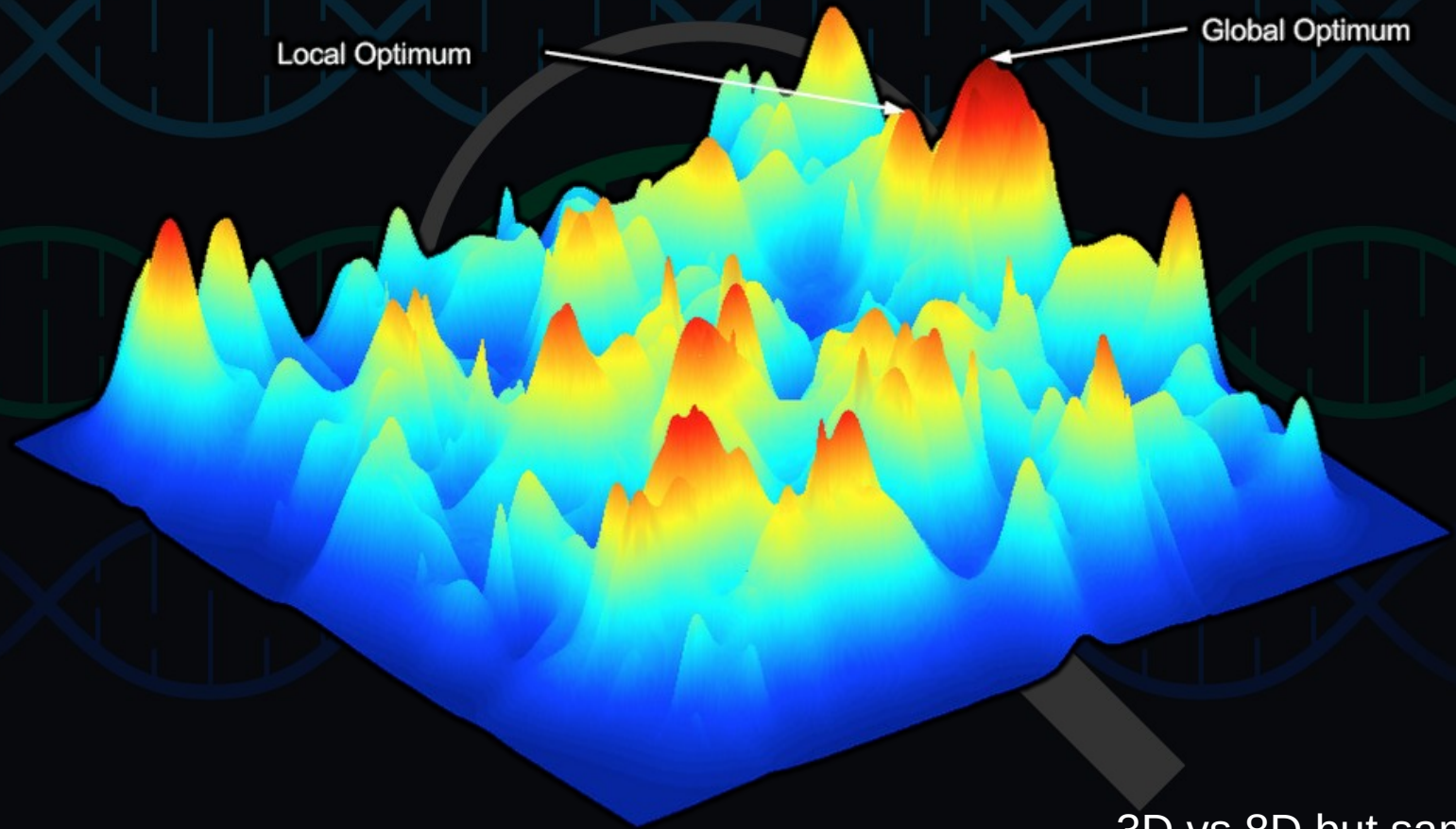
Linear vs Non-Linear networks

$$S(x) = \sum_{i=1}^n w_i x_i + b$$

$$S(x) = \sum_{j_{L-1}} w_{1,j_{L-1}}^{(L)} \max\left(0, \sum_{j_{L-2}} w_{j_{L-1},j_{L-2}}^{(L-1)} \max\left(0, \dots \max\left(0, \sum_i w_{j_1,i}^{(1)} x_i + b_{j_1}^{(1)}\right) \dots + b_{j_{L-1}}^{(L-1)}\right) + b_1^{(L)}\right)$$



Local vs Global Optimum



3D vs 8D but same principle

Global exploration phase

